

HDOC Day-31 Activity

Menu-Driven Operations on a 2D Array

Problem Statement

Write a C program to perform the following menu-driven operations on a given 2D array:

1. Find the sum of all elements in a matrix.
2. Find the sum of each row of a matrix and store it in an array.
3. Find the transpose of a matrix.
4. Exit.

Algorithm

1. Start.
2. Input the number of rows and columns of the matrix.
3. Input all elements of the matrix using nested loops.
4. Display the menu with four choices: sum of all elements, row sums, transpose, and exit.
5. Read the user choice and use switch-case to perform the selected operation.
6. For choice 1, initialize sum = 0 and add every matrix element using nested loops.
7. For choice 2, calculate the sum of each row, store each sum in rowSum[i], and display the row-sum array.
8. For choice 3, display the transpose by interchanging the row and column traversal order.
9. For choice 4, terminate the program.
10. For any other choice, display "Invalid choice".
11. Repeat the menu until choice 4 is selected.
12. Stop.

Test-Case Table

Test Case	Matrix Size	Matrix Elements	Choice	Expected Output	Result
1	2 x 3: 1 2 3 / 4 5 6	1	Sum of all elements = 21	Pass	Pass
2	2 x 3: 1 2 3 / 4 5 6	2	Row sums: 6, 15; array [6, 15]	Pass	Pass
3	2 x 3: 1 2 3 / 4 5 6	3	Transpose: 1 4 / 2 5 / 3 6	Pass	Pass
4	2 x 3: 1 2 3 / 4 5 6	5	Invalid choice	Pass	Pass
5	2 x 3: 1 2 3 / 4 5 6	4	Exiting program...	Pass	Pass

C Program

```
#include <stdio.h>

int main()
{
    int a[10][10], rowSum[10];
    int r, c, i, j, choice, sum;

    printf("Enter the number of rows and columns: ");
    scanf("%d%d", &r, &c);

    printf("Enter the matrix elements:\n");
    for (i = 0; i < r; i++)
    {
        for (j = 0; j < c; j++)
            scanf("%d", &a[i][j]);
    }

    do
    {
        printf("\n1. Sum of all elements\n");
        printf("2. Sum of each row\n");
        printf("3. Transpose of matrix\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                sum = 0;
                for (i = 0; i < r; i++)
                    for (j = 0; j < c; j++)
                        sum = sum + a[i][j];
                printf("Sum of all elements = %d\n", sum);
                break;

            case 2:
                printf("Row sums:\n");
                for (i = 0; i < r; i++)
                {
                    rowSum[i] = 0;
                    for (j = 0; j < c; j++)
                        rowSum[i] = rowSum[i] + a[i][j];
                    printf("Sum of row %d = %d\n", i + 1, rowSum[i]);
                }

                printf("The row-sum array contains: [");
                for (i = 0; i < r; i++)
                {
                    printf("%d", rowSum[i]);
                    if (i < r - 1)
                        printf(", ");
                }
                printf("]\n");
                break;

            case 3:
                printf("Transpose of the matrix:\n");
                for (j = 0; j < c; j++)
                {
                    for (i = 0; i < r; i++)
                        printf("%d\t", a[i][j]);
                    printf("\n");
                }
                break;

            case 4:
                printf("Exiting program...\n");
```

```

        break;

    default:
        printf("Invalid choice\n");
    }
} while (choice != 4);

return 0;
}

```

Compilation

```
gcc day31.c -o day31
```

```

(kali㉿kali)-[~/hdoc31]
$ gcc hdoc31.c -o hdoc31

```

Execution

```
./day31
```

Input: rows = 2, columns = 3; matrix = 1 2 3 / 4 5 6; choice = 1

```

(kali㉿kali)-[~/hdoc31]
$ ./hdoc31
Enter the number of rows and columns: 2 3
Enter the matrix elements:
1 2 3
4 5 6

MENU
1. Sum of all elements
2. Sum of each row
3. Transpose of matrix
4. Exit
Enter your choice: 1
Sum of all elements = 21

```

Input: rows = 2, columns = 3; matrix = 1 2 3 / 4 5 6; choice = 2

```

MENU
1. Sum of all elements
2. Sum of each row
3. Transpose of matrix
4. Exit
Enter your choice: 2
Row sums:
Sum of row 1 = 6
Sum of row 2 = 15
The row-sum array contains: [6, 15]

```

Input: rows = 2, columns = 3; matrix = 1 2 3 / 4 5 6; choice = 3

```

MENU
1. Sum of all elements
2. Sum of each row
3. Transpose of matrix
4. Exit
Enter your choice: 3
Transpose of the matrix:
1      4
2      5
3      6

```

Input: rows = 2, columns = 3; matrix = 1 2 3 / 4 5 6; choice = 5

```
MENU
1. Sum of all elements
2. Sum of each row
3. Transpose of matrix
4. Exit
Enter your choice: 5
Invalid choice
```

Input: rows = 2, columns = 3; matrix = 1 2 3 / 4 5 6; choice = 4

```
MENU
1. Sum of all elements
2. Sum of each row
3. Transpose of matrix
4. Exit
Enter your choice: 4
Exiting program...

(kali@kali)-[~/hdoc31]
$
```

Result

The menu-driven C program for a 2D array was written, compiled, executed, and tested successfully. It correctly finds the sum of all matrix elements, stores and displays the sum of each row in an array, finds the transpose of the matrix, handles invalid menu choices, and exits when the Exit option is selected.